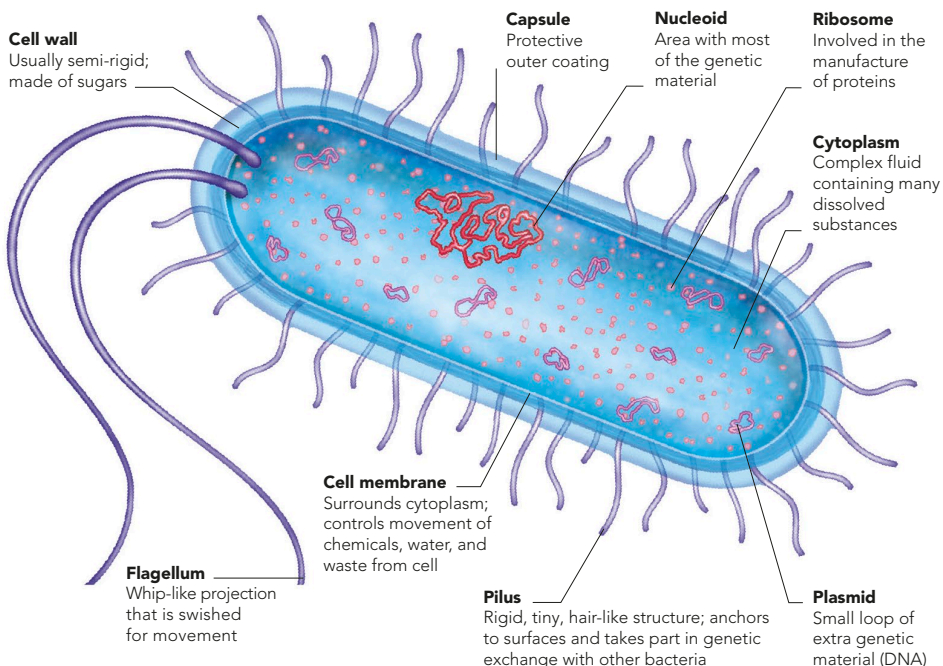


BACTERIA

The microorganisms known as bacteria are present almost everywhere – in soil, water, air, food, drink, and on and in our own bodies. Many types of bacteria are harmless; indeed, those present naturally in the human intestines, the “gut flora”, have a beneficial effect in helping to extract nutrients from food. However, hundreds of types of bacteria can cause infections, ranging from mild to lethal. Bacteria are simpler than other single-cell organisms in that their genetic material (DNA) is free in the cell, rather than contained in a membrane-bound nucleus.

STRUCTURE OF A BACTERIUM

A typical rod-shaped bacterium (bacillus) has a cell membrane enclosing cytoplasm and organelles, such as ribosomes, which are distributed in it. Unlike animal cells, it has a semi-rigid cell wall outside its cell membrane.



BACTERIAL SHAPES

There are several typical shapes for bacteria, and these, along with the way they are coloured by laboratory stains, are important for classification and working out their origins and relationships. Many thousands of bacterial types are known, with more discovered each year.

Cocci

Generally spherical, may exist in clumps, chains, or pairs. Examples include *Staphylococcus* and *Streptococcus*.



Bacilli

Oval, or rod-like, with or without surface hairs or flagella. Examples include *Streptobacillus*.



Spirilla

Spiral or, more accurately, helical (corkscrew-like) in shape. Examples include *Leptospira* and *Treponema*.

